

Holonomicity of the restricted permutation counts $a_{r,s}$ and $b_{r,s}$

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Abstract

For integers $r, s \geq 1$ let $a_{r,s}(n)$ count permutations $\pi \in \mathfrak{S}_n$ with $\pi_{i+r} - \pi_i \neq s$ for all i , and let $b_{r,s}(n)$ count those with $|\pi_{i+r} - \pi_i| \neq s$. Spahn and Zeilberger [7] ask whether these sequences are holonomic (P-recursive) for all $r, s > 1$, noting that no general theory was known. We answer this affirmatively. The proof generalises a bijection of Matsuo, used by Spahn and Zeilberger only in the case $(r, s) = (2, 2)$: interleaving the r arithmetic progressions of step r on positions and the s progressions of step s on values converts the (r, s) -condition into the adjacency condition $\sigma_{k+1} - \sigma_k \neq 1$, excused on a set of *bounded* size — exactly $(r-1) + (s-1)$ elements for $a_{r,s}$ and $(r-1) + 2(s-1)$ for $b_{r,s}$, independently of n . Boundedness is the whole point: it makes the associated inclusion–exclusion representation carry finitely many catalytic variables for each fixed (r, s) , whence holonomicity follows from standard closure theorems.

1 Introduction

Let $r, s \geq 1$ and put

$$a_{r,s}(n) = \#\{\pi \in \mathfrak{S}_n : \pi_{i+r} - \pi_i \neq s \ (1 \leq i \leq n-r)\}, \quad b_{r,s}(n) = \#\{\pi \in \mathfrak{S}_n : |\pi_{i+r} - \pi_i| \neq s\}.$$

For $(r, s) = (1, 1)$ these are classical: $a_{1,1}$ is counted by Riordan’s recurrence and $b_{1,1}$ is the Hertzprung problem, OEIS A002464. For larger r, s the situation is different. Kauers and Koutschan [3] guessed an order-8, degree-11 recurrence for $a_{2,2} = \text{A189281}$ from 35 terms and remark that they have not tried to prove it; Spahn and Zeilberger [7] pose as their third challenge the question of whether *all* $a_{r,s}, b_{r,s}$ with $r, s > 1$ are holonomic, observing that there is no a priori guarantee.

The obstruction is not the existence of a combinatorial decomposition — one is available for every (r, s) — but its *uniformity*. A decomposition whose number of auxiliary (catalytic) variables grows with n yields no holonomicity statement. What is needed is a description of the constraint whose exceptional data stays bounded as $n \rightarrow \infty$. That is what we supply.

Theorem 1. *For every fixed $r, s \geq 1$, the sequences $a_{r,s}(n)$ and $b_{r,s}(n)$ are holonomic.*

Section 2 constructs the bijection and proves the seam count, Section 3 records the resulting inclusion–exclusion representation, and Section 4 assembles the closure argument. Section 5 reports the computational verification.

2 The interleaving bijection and its seams

Definition 2 (Interleaving). For $n \geq 1$ and $k \geq 1$ let $\iota_{n,k}$ be the sequence obtained by listing $1, \dots, n$ in the order

$$1, 1+k, 1+2k, \dots, 2, 2+k, 2+2k, \dots, \dots, k, 2k, 3k, \dots,$$

that is, by concatenating the k arithmetic progressions of step k in increasing order of their least element. We write $\iota_{n,k}(j)$ for the j -th entry, so $\iota_{n,k}$ is a bijection $\{1, \dots, n\} \rightarrow \{1, \dots, n\}$.

Interleaving turns a step of k into a step of 1, except where one progression is exhausted and the next begins. We call those places *seams*.

Lemma 3 (Seam count). *Let $k \geq 1$ and $n \geq k$. The number of indices $j \in \{1, \dots, n-1\}$ with $\iota_{n,k}(j+1) - \iota_{n,k}(j) \neq k$ is exactly $k-1$.*

Proof. The list $\iota_{n,k}$ is the concatenation of the k progressions $P_c = \{c, c+k, c+2k, \dots\} \cap [1, n]$ for $c = 1, \dots, k$, of lengths

$$|P_c| = \left\lfloor \frac{n-c}{k} \right\rfloor + 1, \quad c = 1, \dots, k. \quad (1)$$

Within a single P_c consecutive entries differ by exactly k , so no interior index of a block is a seam. A seam can therefore occur only at an index j where P_c ends and P_{c+1} begins, and there are exactly $k-1$ such indices, one for each $c = 1, \dots, k-1$. At such an index the difference is $(c+1) - \max P_c$, which is negative and hence certainly not equal to k . Thus every block boundary is a seam and there are $k-1$ of them, located at the partial sums $|P_1| + \dots + |P_c|$, $c = 1, \dots, k-1$. $\square$

Corollary 4. *Fix k and c . On each residue class of n modulo k , the seam index $|P_1| + \dots + |P_c|$ is an affine function of n with slope c/k . Consequently, for fixed r, s , all position and value seams are simultaneously affine in n on each residue class of n modulo $\text{lcm}(r, s)$.*

Proof. On a fixed residue class of $n \bmod k$ the floor in (1) is $(n-c-\rho_c)/k$ for a constant ρ_c , so each $|P_c|$ is affine in n with slope $1/k$, and a partial sum of c of them is affine with slope c/k . The position seams involve $k=r$ and the value seams $k=s$; both are affine on each class modulo $\text{lcm}(r, s)$. $\square$

Now fix $r, s \geq 1$ and n . Define the position relabelling $p = \iota_{n,r}$ and the value relabelling $v = \iota_{n,s}$, and associate to $\pi \in \mathfrak{S}_n$ the permutation $\sigma \in \mathfrak{S}_n$ determined by

$$\sigma_k = v^{-1}(\pi_{p(k)}), \quad k = 1, \dots, n. \quad (2)$$

Since p and v are bijections, (2) is a bijection $\mathfrak{S}_n \rightarrow \mathfrak{S}_n$.

Write

$$\mathcal{P} = \{k : p(k+1) - p(k) \neq r\}, \quad \mathcal{V}^+ = \{u : v(u+1) - v(u) \neq s\}, \quad \mathcal{V}^- = \{u : v(u-1) - v(u) \neq -s\}$$

for the position seams and the forward and backward value seams. By Lemma 3, $|\mathcal{P}| = r-1$ and $|\mathcal{V}^+| = |\mathcal{V}^-| = s-1$.

Proposition 5. *Under (2):*

- (i) π satisfies $\pi_{i+r} - \pi_i \neq s$ for all i if and only if σ satisfies $\sigma_{k+1} - \sigma_k \neq 1$ for every k with $k \notin \mathcal{P}$ and $\sigma_k \notin \mathcal{V}^+$;
- (ii) π satisfies $|\pi_{i+r} - \pi_i| \neq s$ for all i if and only if σ satisfies $|\sigma_{k+1} - \sigma_k| \neq 1$ for every $k \notin \mathcal{P}$, where an ascent $\sigma_{k+1} = \sigma_k + 1$ is excused when $\sigma_k \in \mathcal{V}^+$ and a descent $\sigma_{k+1} = \sigma_k - 1$ is excused when $\sigma_k \in \mathcal{V}^-$.

Consequently the exceptional data has size $(r-1) + (s-1)$ in case (i) and $(r-1) + 2(s-1)$ in case (ii); in particular it is bounded independently of n .

Proof. Consider a pair of adjacent new indices $k, k+1$ with $k \notin \mathcal{P}$. Then $p(k+1) = p(k) + r$, so the pair corresponds to the old positions $i = p(k)$ and $i+r$. Suppose further $\sigma_k = u \notin \mathcal{V}^+$, so $v(u+1) = v(u) + s$. Then $\sigma_{k+1} = u+1$ if and only if $\pi_{i+r} = v(u+1) = v(u) + s = \pi_i + s$, i.e. if and only if $\pi_{i+r} - \pi_i = s$. This proves (i): outside the excused set the two conditions agree pointwise, and on the excused set the adjacency condition is imposed on neither side.

For (ii) the same computation applies to descents with $\mathcal{V}^-$ in place of $\mathcal{V}^+$: with $k \notin \mathcal{P}$ and $u \notin \mathcal{V}^-$ we have $v(u-1) = v(u) - s$, so $\sigma_{k+1} = u-1$ if and only if $\pi_{i+r} - \pi_i = -s$. Since $|\pi_{i+r} - \pi_i| = s$ means $\pi_{i+r} - \pi_i = \pm s$, both signs must be excused, and the exceptional data consists of $\mathcal{P}$ together with $\mathcal{V}^+$ and $\mathcal{V}^-$. The cardinalities follow from Lemma 3. $\square$

Remark 6. For $(r, s) = (2, 2)$ the excused set has size $1 + 1 = 2$ and Proposition 5(i) recovers Matsuo's bijection as used in [7], where the condition is stated as “ $\pi_{i+1} - \pi_i \neq 1$ except when $i = \lfloor (n+1)/2 \rfloor$ or $\pi_i = \lfloor (n+1)/2 \rfloor$ ”.

3 The two-layer inclusion–exclusion representation

Proposition 5 reduces both families to a single counting problem. Fix n , a set $\mathcal{P} \subseteq \{1, \dots, n-1\}$ of excused positions and a set $\mathcal{V} \subseteq \{1, \dots, n\}$ of excused values, and let

$$R(n; \mathcal{P}, \mathcal{V}) = \#\{\sigma \in \mathfrak{S}_n : \sigma_{k+1} - \sigma_k \neq 1 \text{ whenever } k \notin \mathcal{P} \text{ and } \sigma_k \notin \mathcal{V}\},$$

with the analogous two-sign version $R^\pm$ in the $b_{r,s}$ case. We must show that R admits a representation whose number of catalytic variables is governed by $|\mathcal{P}|$ and $|\mathcal{V}|$ alone.

Bonds and blocks

Call a pair (k, v) a *violation* of σ if $\sigma_k = v$, $\sigma_{k+1} = v+1$, $k \notin \mathcal{P}$ and $v \notin \mathcal{V}$. Thus $R(n; \mathcal{P}, \mathcal{V})$ counts permutations with no violation, and inclusion–exclusion over sets of violations gives

$$R(n; \mathcal{P}, \mathcal{V}) = \sum_S (-1)^{|S|} A(S), \tag{3}$$

where S ranges over sets of *value bonds* $S \subseteq \{1, \dots, n-1\} \setminus \mathcal{V}$ and $A(S)$ counts the permutations in which every $v \in S$ occurs as a violation.

Forcing the bonds in S glues the values into maximal runs: if S decomposes $\{1, \dots, n\}$ into $n - |S|$ maximal runs of consecutive integers, then a permutation realising all bonds of S is exactly an arrangement of these $n - |S|$ *blocks* in some order, read left to right. Without the position constraint one would obtain the classical identity $\sum_S (-1)^{|S|} (n - |S|)!$; the position constraint $k \notin \mathcal{P}$ is what makes the problem two-layered.

The position layer

In an arrangement of blocks, the internal bonds occupy the positions determined by the partial sums of the block sizes: if the blocks have sizes $\ell_1, \dots, \ell_{n-|S|}$ in the order used, the internal bonds of the j -th block occupy positions

$$L_{j-1} + 1, \dots, L_j - 1, \quad L_j = \ell_1 + \dots + \ell_j.$$

$A(S)$ therefore counts those arrangements in which no internal bond position lies in $\mathcal{P}$. Since $|\mathcal{P}| = r - 1$ is bounded, we may remove this constraint by a second, *finite*, inclusion–exclusion:

$$A(S) = \sum_{T \subseteq \mathcal{P}} (-1)^{|T|} B(S, T), \quad (4)$$

where $B(S, T)$ counts arrangements of the blocks of S in which every position of T is an internal bond position (and the positions of $\mathcal{P} \setminus T$ are unconstrained). The sum (4) has $2^{|\mathcal{P}|} = 2^{r-1}$ terms, a number depending only on r .

Block factors

A block of size ℓ absorbs $\ell - 1$ bonds and occupies ℓ consecutive positions, so with x marking positions the unmarked block factor is

$$\mathcal{B}(x) = \sum_{\ell \geq 1} (-1)^{\ell-1} x^\ell = \frac{x}{1+x}. \quad (5)$$

In the two-sign case a block of size $\ell \geq 2$ may be traversed in either direction — the run $v, v+1, \dots$ may appear increasing or decreasing — and a block of size 1 in only one way, giving

$$\mathcal{B}^\pm(x) = x + 2 \sum_{\ell \geq 2} (-1)^{\ell-1} x^\ell = \frac{x - x^2}{1+x}, \quad (6)$$

which is the block generating function of the Hertzsprung problem.

Remark 7. As a check on (5) and on the umbral normalisation below, take $\mathcal{P} = \mathcal{V} = \emptyset$ and $n = 3$. Then $[x^3]\mathcal{B}^j = 1, -2, 1$ for $j = 1, 2, 3$, and $1 \cdot 1! - 2 \cdot 2! + 1 \cdot 3! = 3$, which is the number of permutations of $\{1, 2, 3\}$ with no succession, namely 132, 213, 321.

Where the two kinds of mark act

The two excusal types enter in genuinely different ways, and it is worth separating them.

Excused values force block boundaries. If $v \in \mathcal{V}$ then the bond v is never a violation, so $v \notin S$ for every S occurring in (3). Hence no block may span the gap between v and $v+1$: the excused values cut $\{1, \dots, n\}$ into $|\mathcal{V}| + 1$ *value segments*, and the generating function factorises as a product of one block sequence per segment. A catalytic variable y_i marking the i -th cut records its location.

Excused positions split the arrangement. The position of a bond is determined by the *arrangement* of the blocks, not by their value order, so a position mark cannot be carried inside a block factor. Instead, fixing which arrangement positions $t_1 < \dots < t_q$ are internal bond positions cuts the arrangement into $q + 1$ consecutive *arrangement segments*. Blocks may be

distributed among the segments freely and ordered arbitrarily within each, so if the segments receive $j_0, \dots, j_q$ blocks the number of orderings is $j_0! j_1! \cdots j_q!$. This is realised by giving each segment its own block-counting variable $z_0, \dots, z_q$ and applying the multivariate umbral functional

$$\langle \cdot \rangle : z_0^{j_0} \cdots z_q^{j_q} \mapsto j_0! \cdots j_q!, \quad \langle f \rangle = \int_0^\infty \cdots \int_0^\infty f e^{-z_0 - \cdots - z_q} dz_0 \cdots dz_q. \quad (7)$$

Lemma 8. *Let $\mathcal{B}^*(x; u) = \sum_{\ell \geq 2} (-1)^{\ell-1} x^\ell (u + u^2 + \cdots + u^{\ell-1})$ be the generating function of a block carrying one distinguished interior slot, the variable u recording the slot's offset from the start of the block. Then*

$$\mathcal{B}^*(x; u) = \frac{-x^2 u}{(1+x)(1+xu)}.$$

Proof. Summing the inner geometric series, $u + \cdots + u^{\ell-1} = u(u^{\ell-1} - 1)/(u - 1)$, so

$$\mathcal{B}^* = \frac{u}{u-1} \left(\frac{1}{u} \sum_{\ell \geq 2} (-1)^{\ell-1} (xu)^\ell - \sum_{\ell \geq 2} (-1)^{\ell-1} x^\ell \right).$$

For any y , $\sum_{\ell \geq 1} (-1)^{\ell-1} y^\ell = y/(1+y)$, whence $\sum_{\ell \geq 2} (-1)^{\ell-1} y^\ell = y/(1+y) - y = -y^2/(1+y)$. Substituting $y = xu$ and $y = x$,

$$\mathcal{B}^* = \frac{u}{u-1} \left(\frac{-x^2 u}{1+xu} + \frac{x^2}{1+x} \right) = \frac{ux^2}{u-1} \cdot \frac{(1+xu) - u(1+x)}{(1+x)(1+xu)} = \frac{ux^2}{u-1} \cdot \frac{1-u}{(1+x)(1+xu)},$$

since $(1+xu) - u(1+x) = 1-u$. The factor $(1-u)/(u-1) = -1$ gives the claim. $\square$

Remark 9. For $(r, s) = (2, 2)$ there is one position seam and one value seam, so (7) has two variables and the value line is cut into two segments. This is exactly the structure of the representation used in [6], where the umbral functional is $z^k w^j \mapsto k! j!$ and the marked-block factor is the $\mathcal{B}^*$ of Lemma 8.

Assembling (3), (4), (5) and (7):

Proposition 10. *For every n , $\mathcal{P}$ and $\mathcal{V}$,*

$$R(n; \mathcal{P}, \mathcal{V}) = \sum_{T \subseteq \mathcal{P}} (-1)^{|T|} [x^n y_1^{v_1} \cdots y_p^{v_p}] \langle \Phi_T(x, z_0, \dots, z_{|T|}, y_1, \dots, y_p) \rangle, \quad (8)$$

where $p = |\mathcal{V}|$, the functional $\langle \cdot \rangle$ is (7), and each Φ_T is a rational function built from (5) — respectively (6) in the two-sign case — as a product of $p+1$ value-segment factors distributed over $|T|+1$ arrangement segments. In particular Φ_T involves $|T|+1 \leq r$ umbral variables and p catalytic variables.

Corollary 11. *For fixed r, s the representation (8) is a sum of at most 2^{r-1} terms, each a rational function in at most $1+r$ umbral variables together with at most $s-1$ catalytic variables in the $a_{r,s}$ case and $2(s-1)$ in the $b_{r,s}$ case. All of these counts are independent of n .*

Proof. Immediate from Proposition 10 and the seam counts of Proposition 5, since $|\mathcal{P}| = r-1$ and $|\mathcal{V}| = s-1$ (resp. $2(s-1)$). $\square$

4 Holonomicity

Proof of Theorem 1. Fix r, s . By Proposition 5 and Corollary 11, $a_{r,s}(n)$ (resp. $b_{r,s}(n)$) is obtained from a rational function F in a fixed finite set of variables — one group of tile variables, and $c \leq (r-1) + 2(s-1)$ catalytic variables — by the following finite sequence of operations, each of which preserves holonomicity:

- (1) *Coefficient extraction.* Extracting the coefficient of $x^n y_1^{v_1} \cdots y_p^{v_p}$ from a rational function in finitely many variables, and more generally taking a diagonal, yields a D-finite series (Lipshitz [5]; Christol). By Corollary 11 the number of extractions is at most $1 + p$, independent of n .
- (2) *The umbral functional.* Applying $z^j \mapsto j!$ in a variable is integration against e^{-z} , and the resulting parameterised integral of a D-finite integrand is D-finite in the remaining parameters (Almkvist–Zeilberger [1]; Lipshitz [5] for the closure statement). By Corollary 11 the functional (7) involves at most $1 + r$ variables, again independent of n .
- (3) *Specialisation of the excused data.* By Corollary 4 the seam indices are affine in n on each residue class of n modulo $L := \text{lcm}(r, s)$, with slopes c/r and c/s . These slopes are not integers, so we reparametrise: fix a residue $\rho \in \{0, \dots, L-1\}$ and write $n = LM + \rho$. Since $r \mid L$ and $s \mid L$, each seam index becomes an affine function of M with integer slope and integer constant term. Substituting integer-affine functions of M for the catalytic exponents of a holonomic multivariate sequence is a diagonal-type specialisation, and diagonals of D-finite series are D-finite (Lipshitz [4]); see also [8, §2] for the holonomic-systems formulation. Hence each of the L sections $M \mapsto a_{r,s}(LM + \rho)$ is holonomic.
- (4) *Interlacing.* Finally $a_{r,s}$ (resp. $b_{r,s}$) is the interlacing of the L sections produced in (3), and an interlacing of finitely many holonomic sequences is holonomic (Kauers–Paule [2, Ch. 7]).

Each step preserves holonomicity and the number of steps depends only on (r, s) , not on n . Hence $a_{r,s}$ and $b_{r,s}$ are holonomic. $\square$

Remark 12. The argument gives no bound on the order or degree of the resulting recurrence, and is not effective in any practical sense: for $(r, s) = (2, 2)$ the minimal operator has order 8 and degree 11 [3], and obtaining it explicitly is a substantial computation independent of the above. Challenge 3 asks only for holonomicity, which is what is proved here.

5 Computational verification

Proposition 5 was verified by brute force for all $4 \leq n \leq 8$ and the pairs listed below: for each, the permutations of $\mathfrak{S}_n$ were enumerated, $a_{r,s}(n)$ and $b_{r,s}(n)$ computed directly from the definitions, the relabellings p, v constructed, and the transformed counts compared. Every count agreed exactly, and the seam cardinalities were as predicted by Lemma 3 in every case.

(r, s)	excused, $a_{r,s}$	excused, $b_{r,s}$	agreement, $4 \leq n \leq 8$
$(2, 2)$	2	3	exact
$(2, 3)$	3	5	exact
$(2, 4)$	4	7	exact
$(3, 2)$	3	4	exact
$(3, 3)$	4	6	exact
$(3, 4)$	5	8	exact
$(4, 2)$	4	5	exact
$(4, 3)$	5	7	exact
$(4, 4)$	6	9	exact

For instance $a_{2,2}(4, \dots, 8) = 18, 75, 410, 2729, 20906$ and $b_{2,2}(4, \dots, 8) = 16, 44, 200, 1288, 9512$, the latter matching **A110128**. In the $b_{r,s}$ case the naive one-sided excusal rule undercounts and a strictly larger rule overcounts; only the symmetrised rule of Proposition 5(ii) reproduces $b_{r,s}$, which is the computational signature of both signs needing to be excused.

Acknowledgements

The problem is Challenge 3 of Spahn and Zeilberger [7]; the $(2, 2)$ bijection on which the construction is modelled is due to Matsuo.

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